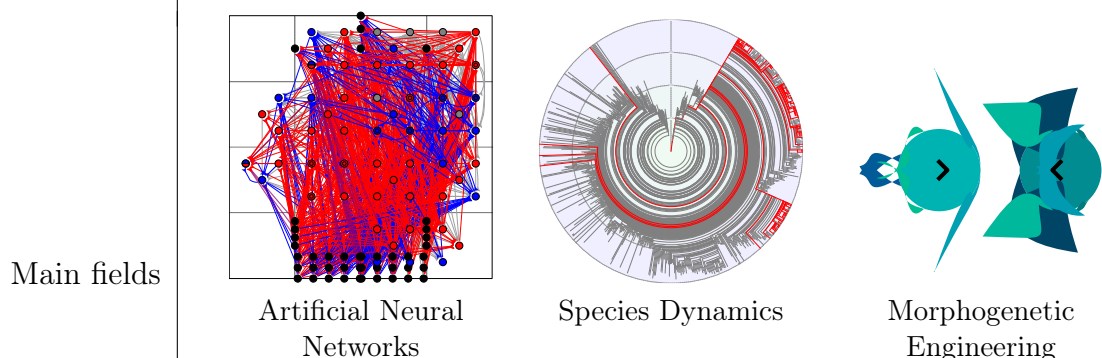


Dr. Kevin Godin-Dubois

Contact	✉ k.j.m.godin-dubois@vu.nl	🌐 kgd-al.github.io
	🏠 Vrije Universiteit Amsterdam de Boelelaan 1081a, 1081HV Amsterdam, The Netherlands	📧 kgd-al@github.com 🐦 godinduboisalife 🆔 0009-0002-6033-3555
Position	Researcher in Evolutionary Robotics (since November 2022)	

Highlights

Research *Artificial Life: Morphology, Cognition & Interaction*



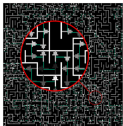
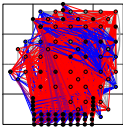
Output 3 journal articles (*Artificial Life*, *Frontiers*, *JOSS*)
5 international conference articles (*ALife*, *IEEE ALife*, *EvoAPP*)
7 international workshops short papers (*ALife*, *ECAL*, *AAAI*)
Scientific software: ABrain, AMaze, Revolve2, SHARPIE

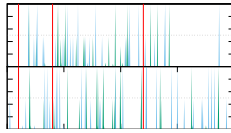
Positions

Postdoctoral 2022 - Present	Computer Science / Evolutionary Robotics “ <i>NeuroEvolution and Reinforcement Learning for Embodied Robots</i> ” Computational Intelligence Group - Vrije Universiteit Amsterdam, The Netherlands Supervisor: Dr. A. Yaman (a.yaman@vu.nl) Collaborators: Dr. A. Kononova (a.kononova@liacs.leidenuniv.nl)
Postdoctoral 2020 - 2022	Computer Science / Artificial Intelligence “ <i>Emergent cognitive architectures in virtual embodied robots</i> ” REVA Team, IRIT - Toulouse I University, France Supervisors: Pr. Y. Duthen (yves.duthen@irit.fr) Pr. S. Cussat-Blanc (sylvain.cussat-blanc@irit.fr)
PhD 2016-2020	Computer Science / Artificial Life “ <i>Environment-driven speciation: long term interactions in artificial plant communities</i> ” REVA Team, IRIT - Toulouse I University, France Supervisors: Pr. Y. Duthen (yves.duthen@irit.fr) Pr. S. Cussat-Blanc (sylvain.cussat-blanc@irit.fr)

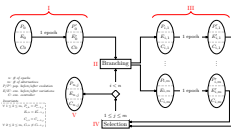
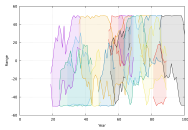
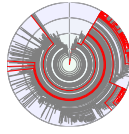
Teaching	9 years (600+ hours)
Computer Science	Learning Machines, Bachelor and Master research projects Programming languages: Python, C, R Algorithms, Data Structures, Information theory Programming projects
Generalists	Data Science tools and languages Database modeling, SQL
Skills	
Programming	Fluent: C++, Bash, Python, L ^A T _E X Working Knowledge: C, Java, R, VB, VBA, JavaScript
Technical	Evolutionary Algorithms, Machine Learning, Multi-Agents Systems, High-Performance Computing
Languages	French (native), English

Research

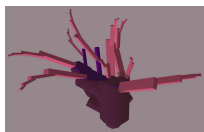
Synopsis	My main interests revolve around autonomous artificial life forms: from the design of efficient morphologies to the emergence of high-level control schemes and the evolutionary constraints that favor both. Recently I am mostly focused on Artificial Neural Networks (ANN) through NeuroEvolution and Reinforcement Learning, notably in the context of Interactive Evolutionary Robotics.
Reinforcement Learning	Complementing Neuro-Evolutionary approaches with Reinforcement-based learning for more robust agents.
	[1, 2] AMAZE Benchmark generator for lightweight sighted agents. Meant as a fast prototyping platform for Reinforcement Learning and NeuroEvolutionary algorithms. Software: AMaze
	[13, 12] SHARPIE Collaborative Hybrid Intelligence project aiming to provide the new standard for interactive experiments in Reinforcement Learning settings. Software: SHARPIE
Artificial Neural Networks	Studying the emergence of various “cognitive” capabilities in virtual robots, controlled by a spontaneously differentiated neural network, in response to biologically plausible stimuli.
	[8, 3, 7] VIRTUAL FMRI Extracting stimulus-specific regions of an ANN by applying a virtual equivalent to functional Magnetic Resonance Imaging (fMRI) and building high-level cognitive maps. Software: ES-HyperNEAT (Custom implementation)



Species Dynamics



Morphogenetic Engineering



Expertise

Evolutionary Algorithms

Machine Learning

Physics simulators

[16, 15] COMMUNICATION

Exploring the mechanisms leading to emergent communication, how it becomes structured and its neural implementation.

Promoting complex evolutionary trajectories and extracting species-level information from individual reproductions.

[19, 17] PHYLOGENETICS

Automatically transforming genealogic trees into phylogenetic abstraction to access the emergent species-level dynamics.

Software: APOGeT(Automated Phylogeny Over Geological Timescales)

[11] SPECIATION

Application of a bio-inspired reproduction operator (Bail-Out Crossover) capable of spontaneously generating species barriers thereby allowing for emergent speciation.

[9, 24] EVOLUTIONARY ALGORITHMS

Introduced a novel paradigm, EDEnS (Environment-Driven Evolutionary Selection), relying on the indirect controlling of whole populations' evolutionary trajectories through an evolvable environmental controller.

Developing functional morphology in response to environmental constraints and evolutionary pressures.

[18, 9, 10, 19] DEVELOPMENTAL MORPHOLOGIES

Production of mature, functional virtual plants from a single cell/structure using various genetic encodings (rules-based, L-Systems, GraphTals) in response to environmental constraints.

[3] VIRTUAL ROBOTS

Use of genetically parameterized cubic bézier curves to control both static and mobile structures on the perimeter of virtual circular robots.

Software: Splinoids

Videos: on Vimeo

- Environment-Driven Evolutionary Selection (EDEnS)

- Multi-objective Optimisation

- High Performance Computing (HPC), Co-evolution, Novelty

- Artificial Neural Networks (ANN, CNN, RNN)

- Composite Pattern-Producing Networks (CPPN)

- Cartesian Genetic Programming (CGP)

- Genetic Regulatory Networks (GRN)

- Hidden Markov Models (HMM)

- Stable baselines 3

- Approximate: Bullet3, Box2D

- Exact: MuJoCo

Teachings

Postdoc 2023-2024	Vrije Universiteit Amsterdam	
	• NeuroEvolution (lecture)	
	• Learning Machines (projects)	45h
	• Master and Bachelor thesis supervision	>100h
Course management 2021-2022	Toulouse I University & Toulouse III University, France	
	• Computer Science projects	72h
	<i>Multi-Agent Systems, Complex Systems, Simulation</i>	
	• R programming	67.5h
	<i>English lectures</i>	
	• Information theory	22.5h
	• Servers and contents	18.75h
Teaching fellow 2017-2021	Toulouse I University & Toulouse III University, France	
	• Statistical software (R & Python)	36h
	• Algorithms	60h
	• Excel & VBA	60h
	• Modeling in databases	21h
Practical work supervisor 2016-2021	Toulouse III University, France	
	• Software projects	69.2h
	• Data structures	18.8h
	• C Programming	36h
	• Python	8h

Outreach

Reviewer 2023	• Symposium on Artificial Life program comitee member	
	• Journal of Open Source Software reviewer	
EduMix Aspi-Friendly 2021	Initiated a project for the self-monitoring of well-being in students with autistic disorders alongside a heterogeneous team of neuro-(a)typical and various profiles (faculty, designers, developers ...).	

Internships

Morphogenetic Engineering 2016 (6 months)	Toulouse Research Institute on Computer Science (IRIT), France <i>“Rule-based artificial embryogenesis in a complex 3D environment”</i> Deployed rule-based genomes on the MecaCell platform to study artificial plant growth and cell specialization. Contact: Pr. Y. Duthen (yves.duthen@irit.fr)
Machine Learning 2015 (3 months)	IRIT, <i>“Comparison of different evolutionary approaches, an application to the GECCO 2015 challenge”</i> Performed a performance comparison (accuracy, efficiency) between Artificial Neural and Genetic Regulatory Networks on the 2015 GECCO temperature prediction challenge data. Contact: Pr. H. Luga (herve.luga@irit.fr)

Machine Learning 2014 (2 months)	IRIT, “ <i>An architecture for automated bird discrimination</i> ” Applied Hidden Markov Models to the BirdClef2014 challenge on the identification of specific bird species in a corpus of thousands of recordings. Contact: Pr. J. Farinas (jerome.farinas@irit.fr)
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Education

PhD 2016 - 2020	Toulouse I University, France Defended the 15th of July 2020 Thesis title: “ <i>Environment-driven speciation: long term interactions in artificial plant communities</i> ” Investigated how complexification of artificial creatures could be further enhanced through the indirect control provided by a co-evolved, highly dynamical environment. Rapporteurs: Pr. P. Collet & DoR. F. Vico Contact: Pr. Y. Duthen (yves.duthen@irit.fr)
Master 2014 - 2016	Toulouse III University, France (<i>with honours</i>) Artificial Intelligence: mathematical & symbolic models, training methods
Bachelor 2011 - 2014	Toulouse III University (<i>with distinction</i>) Computer Science: networks, programming, systems, mathematics

Scholarships and Fellowships

2023-2026 ~ 200K €	Postdoctoral funding from the Hybrid Intelligence consortium (Netherlands)
2016-2019 70K €	PhD Fellowship from the French Minister of Higher Education and Research (MESR)
2015 10K €	Master Scholarship from the International Mathematics and Computer Science Center (LabEx CIMI, Toulouse)
2014-2015 3K6 €	Merit Scholarship from the Regional Student Welfare Office (CROUS, Toulouse)

Journals (peer-reviewed)

- [1] **K. Godin-Dubois**, K. Miras, and Anna V Kononova. “AMaze: An Intuitive Benchmark Generator for Fast Prototyping of Generalizable Agents”. In: *Frontiers in Artificial Intelligence* Volume 8 - 2025 (2025). ISSN: 2624-8212. DOI: 10.3389/frai.2025.1511712.
- [2] **K. Godin-Dubois**, K. Miras, and A. V. Kononova. “AMaze: A Benchmark Generator for Sighted Maze-Navigating Agents”. In: *Journal of Open Source Software* (2025). DOI: 10.21105/joss.07208.
- [3] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Explaining the Neuroevolution of Fighting Creatures Through Virtual fMRI”. In: *Artificial Life* 29.1 (2023), pp. 66–93. ISSN: 1064-5462. DOI: 10.1162/artl_a_00389.

International conferences (peer-reviewed)

- [4] **K. Godin-Dubois**, A. Yaman, and A. V. Kononova. “Benefits of Low-Cost Bio-Inspiration in the Age of Overparametrization”. In: *Parallel Problem Solving From Nature*. 19. Trento, Italy: arXiv, 2026, under review. DOI: 10.48550/ARXIV.2604.20365. (Visited on 04/23/2026).
- [5] L. Goncalvez Braz, F. Den Hengst, **K. Godin-Dubois**, H. Aydin, K. Baraka, S. Wang, and F. A. Oliehoek. “SHARPIE: A Robust Framework for Streamlining Human-AI Reinforcement Learning Experiments”. In: *Reinforcement Learning Conference*. 2026, under review.
- [6] O. Moulin, D. Pasero, L. Bierling, and **K. Godin-Dubois**. “On the Impact of Morphological Encoding for Evolutionary Robotics in ARIEL”. In: *Parallel Problem Solving From Nature*. 19. Trento, Italy, 2026, under review.
- [7] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Specialization or Generalization: Investigating NeuroEvolutionary Choices via Virtual fMRI”. In: *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*. MIT Press, July 2024. DOI: 10.1162/isal_a_00817.
- [8] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Spontaneous Modular NeuroEvolution Arising from a Life/Dinner Paradox”. In: *The 2021 Conference on Artificial Life*. Cambridge, MA: MIT Press, 2021, p. 95. DOI: 10.1162/isal_a_00431. Presentation: <https://vimeo.com/godinduboisalife/alife2021main>.
- [9] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Beneficial Catastrophes: Leveraging Abiotic Constraints through Environment-Driven Evolutionary Selection”. In: *2020 IEEE Symposium Series on Computational Intelligence (SSCI)*. 2020, pp. 94–101. DOI: 10.1109/SSCI47803.2020.9308411.
- [10] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Self-Sustainability Challenges of Plants Colonization Strategies in Virtual 3D Environments”. In: *Applications of Evolutionary Computation*. Ed. by P. Kaufmann and P. A. Castillo. Cham: Springer International Publishing, 2019, pp. 377–392. ISBN: 978-3-030-16692-2. DOI: 10.1007/978-3-030-16692-2_25.
- [11] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Speciation under Changing Environments”. In: *ALIFE 19*. Vol. 31. Cambridge, MA: MIT Press, 2019, pp. 349–356. ISBN: 978-0-262-35844-6. DOI: 10.1162/isal_a_00186. Presentation: <https://vimeo.com/godinduboisalife/alife2019>.

Workshops

- [12] H. Aydin, **K. Godin-Dubois**, L. G. Braz, F. Den Hengst, K. Baraka, M. M. Celikok, A. Sauter, S. Wang, and F. A. Oliehoek. “SHARPIE: A Modular Framework for Reinforcement Learning and Human-AI Interaction Experiments”. In: *Collaborative AI and Modeling of Humans*. arXiv, Feb. 2025. DOI: 10.48550/arXiv.2501.19245. (Visited on 03/14/2025).
- [13] H. Aydin, **K. Godin-Dubois**, L. Goncalvez Braz, Floris Den Hengst, K. Baraka, M. M. c Celikok, A. Sauter, S. Wang, and F. A. Oliehoek. “Towards an Experimentation Platform for Hybrid Human-AI Sequential Decision-Making”. In: *Frontiers in Artificial Intelligence and Applications*. Ed. by D. Pedreschi, M. Milano, I. Tiddi, S. Russell, C. Boldrini, L. Pappalardo, A. Passerini, and S. Wang. IOS Press, Sept. 2025. ISBN: 978-1-64368-611-0. DOI: 10.3233/FAIA250669.
- [14] **K. Godin-Dubois**, O. Weissl, K. Miras, and A. V. Kononova. “Interactive Embodied Evolution for Socially Adept Artificial General Creatures”. In: *Evolution of Things Workshop at the ALife 2024 Conference*. arXiv, July 2024. DOI: 10.48550/arXiv.2407.21357.
- [15] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Emergent Communication for Coordination in Teams of Embodied Agents”. In: *4th International Workshop on Agent-Based Modelling of Human Behaviour (ALife2022)*. 2022.
- [16] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “On the Benefits of Emergent Communication for Threat Appraisal”. In: *3rd International Workshop on Agent-Based Modelling of Human Behaviour*. Online, 2021. Presentation: <https://vimeo.com/godinduboisalife/abmhub2021>.
- [17] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “APOGeT: Automated Phylogeny Over Geological Timescales”. In: *MethAL Workshop at ALife 2019*. 2019. DOI: 10.48550/arXiv.2407.21412.
- [18] **K. Dubois**, S. Cussat-Blanc, and Y. Duthen. “Towards an Artificial Polytrophic Ecosystem”. In: *Morphogenetic Engineering Workshop, at the European Conference on Artificial Life (ECAL) 2017 September 4*. 2017.

Posters

- [19] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Studying Long Term Interactions between Plants and Their Environment”. In: *Alife 2018*. Tokyo, 2018. DOI: 10.13140/RG.2.2.27553.97125.

Oral presentations

- [20] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. *Splinoids: First Steps out of EDEnS*. Talk. Montreal (Virtual), 2020.

Softwares and datasets

- [21] **K. Godin-Dubois**. *AMaze: Fully Discrete Training with Three Regimes (Direct, Scaffolding, Interactive) and Two Algorithms (A2C, PPO)*. Dataset. Feb. 2024. DOI: 10.5281/ZENODO.10622913. (Visited on 09/19/2024).
- [22] **K. Godin-Dubois**, K. Miras, and A. V. Kononova. *AMaze: A Lightweight Benchmark Generator for Sighted Agents*. Zenodo. Software. Apr. 2024. DOI: 10.5281/ZENODO.10907939.

- [23] A. Stuurman, O. Weissl, T.-C. Chiang, AndresG, D. Zeeuwe, **K. Godin-Dubois**, and Roy. *Ci-Group/Revolve2: 1.2.3*. Zenodo. Software. Nov. 2024. DOI: 10.5281/ZENODO.14143431. (Visited on 02/12/2025).

Thesis

- [24] **K. Godin-Dubois**. “Environment-Driven Speciation: Long-Term Interactions in Artificial Plant Communities”. PhD thesis. Doctoral school of Mathematics, Computer Science and Telecommunications (Toulouse, France), 2020.